Module 6

1. Explore Language-Oriented Case Study & Challenges

Case Study 1: Patient Urgency Analysis

This case study focuses on identifying the urgency level of patients based on extracted medical information. The urgency levels are categorized into different classes such as low, medium, and high. A count-based visualization is used to analyse how frequently each urgency level occurs in the dataset.

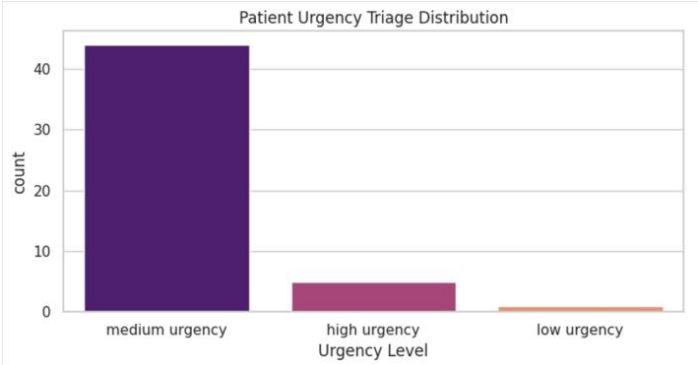

Fig 6.1: Patient Urgency Triage Distribution

Observation:

The distribution helps in understanding how many patients require immediate attention compared to those with less critical conditions. This type of analysis can assist in hospital triage systems and prioritization.

Case Study 2: Extraction of Actionable Instructions

In this case study, the number of actionable instructions extracted from each medical report is analysed. Instructions may include dosage details, precautions, or follow-up steps. A histogram is used to visualize the number of instructions per report.

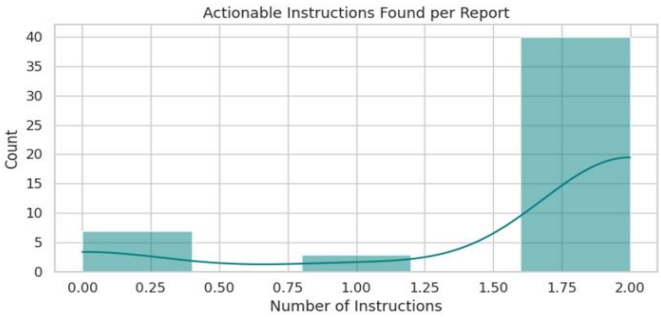

Fig 6.2: Actionable Instructions Found per report

Observation:

Most reports contain a limited number of instructions, while some contain multiple detailed steps. This indicates variability in prescription complexity and helps in evaluating how well the NLP system captures important instructions.

Case Study 3: Clinical Fact Verification (Negation Detection)

This case study examines whether certain medical conditions, such as fever, are present or explicitly negated in the text.

A count plot is used to compare:

- Presence of condition
- Absence or negation

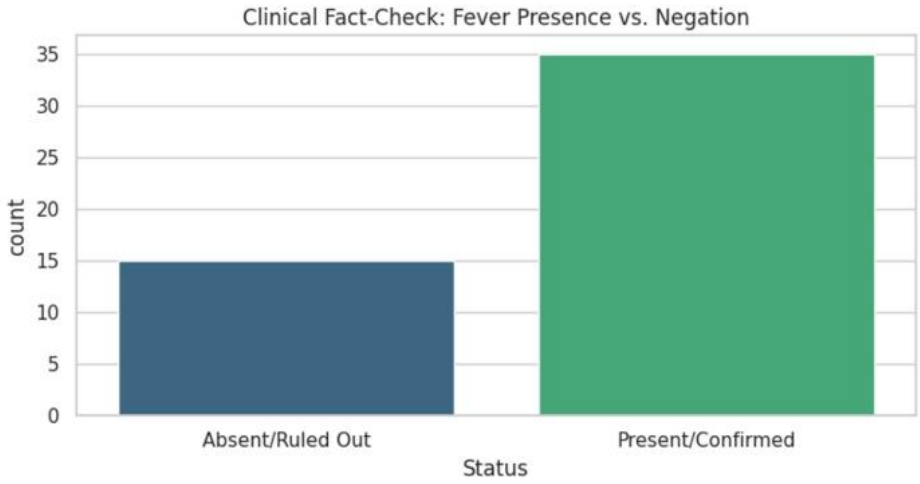

Fig 6.3: Clinical Fact Check

Observation:

Handling negation correctly is crucial in medical NLP. For example, “no fever” should not be interpreted as the presence of fever. This analysis ensures that the system correctly distinguishes such cases.

Case Study 4: Medication Information Complexity

This analysis focuses on the complexity of extracted medication and dosage information. The length of the extracted text is used as a measure of complexity. A box plot is used to visualize variations in the length of medication-related data.

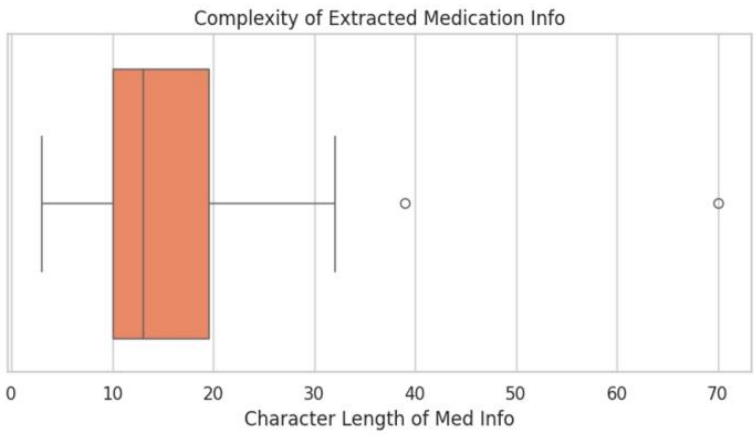

Fig 6.4: Complexity of Extracted Medication Info

Observation:

Some prescriptions contain short and simple medication details, while others are lengthy and complex. This variation reflects the diversity of real-world medical data and highlights the need for robust extraction techniques.

Case Study 5: Emotional State vs Medical Urgency

This case study explores the relationship between a patient's emotional condition and their medical urgency level.

A grouped count plot is used to compare emotional states (such as stress, anxiety, or calmness) with urgency categories.

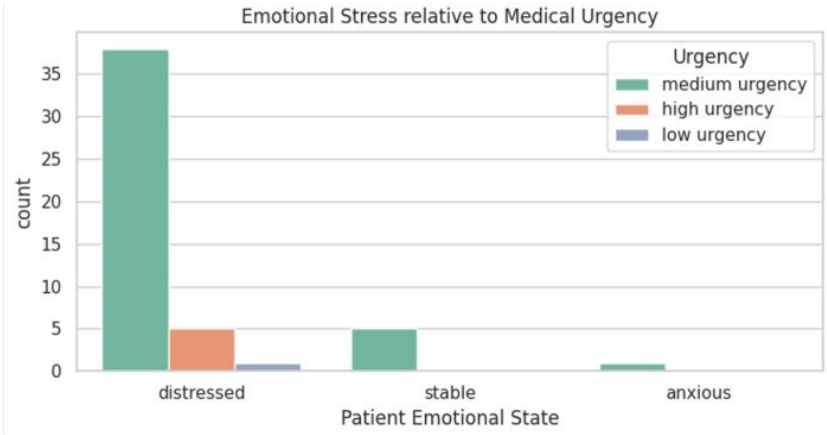

Fig 6.5: Complexity of Extracted Medication Info

2. Apply Machine Learning Techniques using Pretrained Transformers

1. Medical Specialty Classification

The Problem:

Large hospitals generate thousands of unstructured documents daily that must be routed to specific departments for archiving, billing, and resource allocation.

Solution:

Manual sorting is slow, prone to human error, and creates massive administrative backlogs that delay patient care and insurance claims.

Transformer : BERT

It uses a Discriminative Architecture (Encoder-only) that converts complex medical narratives into a single "Context Vector" used to predict a category with high accuracy.

Characteristic:

Bi-directional Contextualization.

Distinguish between a "Cardiology" report and a "Urology" even if they use similar words like "vessel" or "blockage."

2.Clinical Document Summarization

The Problem:

Medical transcriptions are often long, repetitive, and filled with "filler" conversation, making it difficult for specialists to quickly find the "Chief Complaint" or "Final Diagnosis."

Solution:

Doctors suffer from "Information Overload." Distilling a 2,000-word transcription into a 5-sentence summary saves critical time during shift changes.

Transformer : BART

Encoder-Decoder models. The Encoder "compresses" the transcription, and the Decoder "reconstructs" it into a concise, grammatically correct summary.

Characteristic:

Global Self-Attention.

This allows the model to link a symptom mentioned at the beginning of a document to a surgery performed at the end,

3. Named Entity Recognition (NER)

Problem:
Identifying and extracting specific "Variables" (drug names, dosages, frequencies, and anatomical sites) from a wall of narrative text.

Solution : For clinical trials or safety audits, researchers need "Structured Data." It is impossible to manually highlight every medication mentioned across 5,000 files.

Transformer RoBERTa:
It provides High-Resolution Token Representation, meaning it treats every word as a unique entity shaped by its immediate neighbors.

Characteristic:
Dynamic Masking.
During training, the model learns to "guess" missing medical terms, making it incredibly precise at identifying rare drug names or complex chemical compounds it hasn't seen before.

4. Medical Question Answering (QA)
Problem:
Clinicians often need a specific fact (e.g., "What was the heart rate after the second dose?") buried deep within a patient's historical records.
Solution:
Modern "Electronic Health Records" (EHR) are fragmented. A QA system acts as a search engine that understands medical logic rather than just matching keywords.
Transformer DistilBERT: It is optimized for Span Prediction. It scans the transcription and highlights the exact start and end of the answer without needing to rewrite the text.
Characteristic: Distilled Knowledge. It offers nearly the same accuracy as larger models but with a 40% smaller size, allowing it to run instantly on low-power hospital tablets or mobile devices.

5. Clinical Text Generation (Autocompletion)

Problem: Physicians spend more time typing reports than seeing patients. Standard templates are too rigid, but writing from scratch is too slow.

Solution: Implementing an "AI Scribe" that suggests the next sentence helps maintain a natural flow of documentation while adhering to medical standards.

Transformer GPT

These are Decoder-only (Autoregressive) models designed specifically to predict the most statistically likely "next word" based on the previous context.

Characteristic: Causal Language Modeling. The model is trained to look only at "past" words to predict "future" words, enabling it to generate human-like clinical narratives that feel natural and professional.

OUTPUT

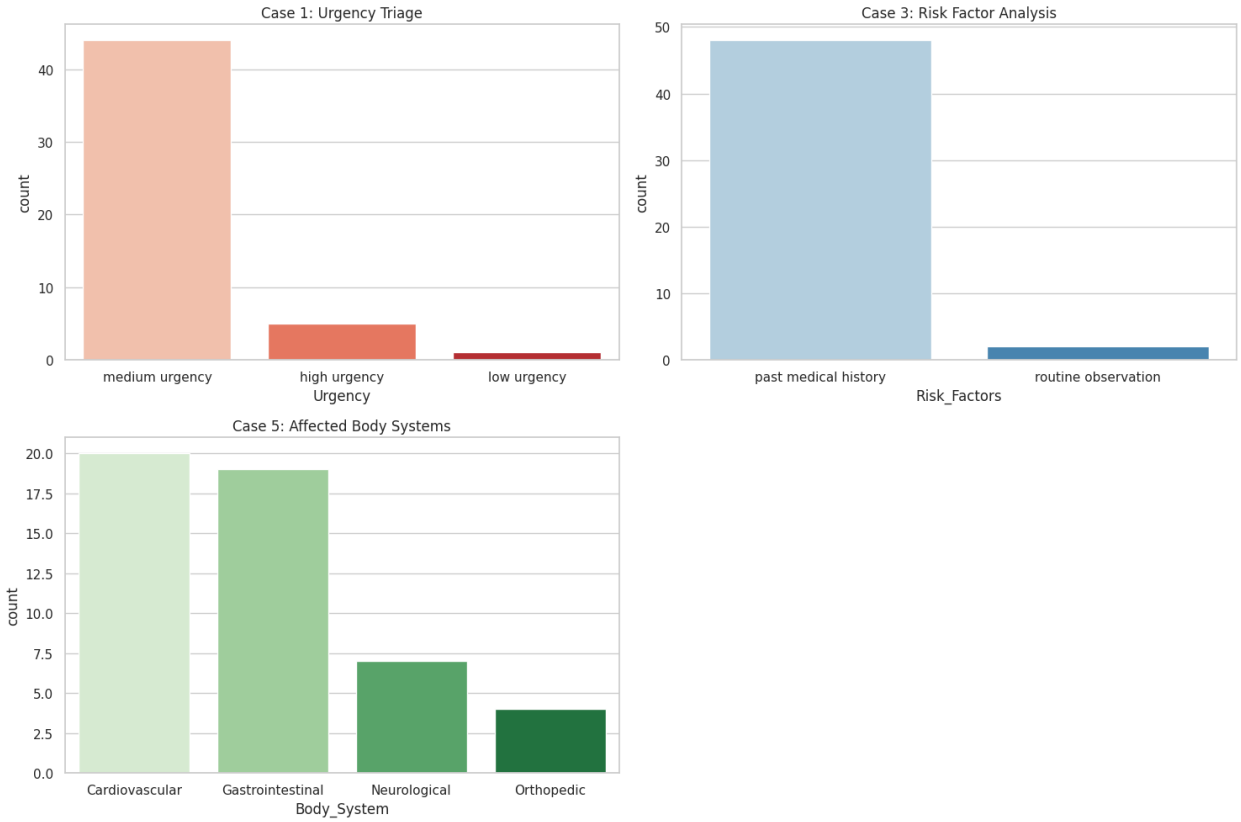

Fig 6.6: Patient Urgency Triage Distribution

	Urgency	Fever_Check	Meds_Dosage	Emotional_Status	Instructions
0	medium urgency	Absent/Ruled Out	Ortho Tri-Cyclen and the Allegra	distressed	[In the past, she has tried Claritin, and Zyrt...
1	medium urgency	Present/Confirmed	Penicillin	distressed	[]
2	medium urgency	Present/Confirmed	alcohol	distressed	[He has asthma and difficulty walking two bloc...
3	medium urgency	Absent/Ruled Out	Mild mitral and tricuspid regurgitation	distressed	[2-D M-MODE: , ,1.]
4	medium urgency	Absent/Ruled Out	1. The left ventricular cavity size and wall ...	distressed	[]
5	medium urgency	Present/Confirmed	general anesthesia	distressed	[PREOPERATIVE DIAGNOSIS: , Morbid obesity.,POS...
6	medium urgency	Present/Confirmed	Lipodystrophy of the abdomen	distressed	[PREOPERATIVE DIAGNOSES:,1., Lipodystrophy of ...
7	medium urgency	Absent/Ruled Out	2-D ECHOCARDIOGRAM	stable	[Cardiac function is normal., Mitral inflow an...
8	medium urgency	Present/Confirmed	Lipodystrophy	distressed	[PREOPERATIVE DIAGNOSIS: , Lipodystrophy of th...
9	medium urgency	Absent/Ruled Out	Trace mitral regurgitation	stable	[Aortic valve seen with good motion.,5., Mitra...

Fig 6.7: Patient Urgency Triage Distribution

Outcome

1. Analyze real-world language problems
2. Apply pretrained transformer models
3. Build end-to-end NLP classification system

RESULT

Hence Case studies and advance ML techniques have been implemented successfully.